

Computer Programmable electronic machine which working under a given set of instruction.

Computer Programming Language Programming Language is a language designed to communicate with computer.

Components of a Language

- Semantics – Meaning
- Syntax – How the meaning is presented.

Generation of Programming Languages

- ***First-Generation Languages :***
 - These are low-level languages like machine language.
- ***Second-Generation Languages :***
 - These are low-level assembly languages used in kernels and hardware drives.
- ***Third-Generation Languages :***
 - These are high-level languages like C, C++, Java, Visual Basic, and JavaScript.
- ***Fourth Generation Languages :***
 - These are languages that consist of statements that are similar to statements in the human language. These are used mainly in database programming and scripting, like Perl, Python, Ruby, SQL, and MatLab
- ***Fifth Generation Languages :***
 - These are the programming languages that have visual tools to develop a program like, Mercury, OPS5, and Prolog.

Translation

• ***Interpreting***

Instructions are translated and executed one after one. HTML,

• ***Compiling***

- Translate all the instructions of the program in a one step and create an executable file.
- Some languages create intermediate form known byte code (java)

Translators

Interpreter

- converts source code statements into equivalent machine language statements and execute.
- Source program is required for every execution
- Slow execution.

Compiler

- Once all errors have been corrected translates the entire source code as a single unit.
- After the compilation process the source program is no more required
- Faster execution.

How Java is working

• ***Source Code***

- Source code is plain text
- File name extension is .java
- Compiler is Javac
- Object code/byte code extension is .class

Virtual machine is a software that emulate a computer system, which provide functionality of a physical computer.

Features of Java Language

- ***Case sensitive*** – Compiler recognize the deferent between simple and capital letters
- ***Secure*** – When this byte codes are executed , the JVM can take care of the security.
- ***Robust*** – Strong memory management – Exception handling mechanism – Type checking mechanism
- ***Portable*** – Program written in Java can be run on deferent computing environments.
- ***Multithreaded*** – Java supports parallel processing through execution of multiple threads.

Comments Statements which are ignored by compiler .

Line comments

// Comments a line of code

Block comments

/* */ Comments a block of code

Integrated Development Environment A software application that helps programmers develop software code efficiently.

Features of an IDE

- **Code editor** – Write and edit code
- **Build automation**
 - Compiling computer source code into binary code
 - Packaging binary code
 - Running automated tests.
- Form Designer** – Allows a programmer to build forms with a drag and drop interface.
- **Debugger**
 - Test and find bugs (errors)
- **Refactoring**
 - Restructuring existing computer code without changing its external behavior.

Popular Java IDE

- NetBeans
- BlueJ
- Eclipse
- Intel J Idea
- JDR java

Features of NetBeans

- Easy to install with easy configurations
- Code Editing.
- Graphical User Interface.
- Debugging
- Form designer

Debugging The process of identifying and removing errors from computer hardware or software.

Types of errors

- **Syntax errors**
 - Syntax errors are detected at compile-time
 - Intelligent IDE identify syntax errors in coding time
- **Logical errors**
 - Error in code logic to produce an unexpected output
 - **Run time error**

Software versioning is the process of assigning either unique version names or unique version numbers to unique states of computer software.

Initialize a repository allows developer to save versions of the code, which can access when needed.

Commit means snapshots or milestones along the timeline of a Git project.

Variable is a location in computer memory which can hold data.

Terminology

- **Key word** – special reserved word which is having a meaning to compiler. So key words cant be used for identifiers
- **Identifiers** - word used by a programmer to name a variable, method, class or label.

Data type Classification of data (value of a variable) which tells the compiler or interpreter how the programmer intends to use the data.

Primitive data types

- **long:** – 64-bit signed two's complement integer – -9,223,372,036,854,775,808 to 9,223,372,036,854,775,807 (inclusive).
- **float:** – Single precision 32-bit IEEE 754 floating point.
- **double:** – Double precision 64-bit IEEE 754 floating point.
- **boolean:** – Two possible values: true and false
- **char:** – Single 16-bit Unicode character – '\u0000' (or 0) to '\uffff' (or 65,535 inclusive).
- **byte:** – 8-bit signed two's complement integer – -128 to 127 (inclusive)
- **short:** – 16-bit signed two's complement integer – -32,768 to 32,767 (inclusive)
- **int:** – 32-bit signed two's complement integer – -2,147,483,648 to 2,147,483,647 (inclusive)

Creating and Accessing Variables

- int A;
- A=10;
- char B;
- B=57;

Variable/Identifier Naming

- Java keywords cant be used as variable names
- Variable names are case sensitive
 - legal identifiers:
- age, \$salary, _value, __1_value
- Illegal identifiers:
- 123abc, -salary.

Variable declaration Informing the compiler about the variables that a program is going to use is variable declaration.

Variable Initialization Setting an initial value to a variable.

Literal The value stored in a variable.

Types of Variables

- **Local Variable** – A variable that is declared inside the method is called local variable.
- **Instance Variable** – A variable that is declared inside the class but outside the method is called instance variable. It is not declared as static. So values of Instance Variable are unique to each instance of a class
- **Static variable** – A variable that is declared as static is called static variable. It cannot be local. There is exactly one copy of this variable, regardless of how many times the class has been instantiated
- **Parameters** – Variables which pass values with method calls.

Type conversion / Type casting Changing a value from one data type to another type is known as data type conversion

• Implicit conversion

– Automatic conversion by JVM

• Explicit conversion

– Not automatically done by JVM

– Programmer need to specify the target type in parentheses.

Widening Conversion Value of lower size data type converted to a value of a higher size data type without loss of information.

Narrowing Conversion value of higher size data type converted to a value of a lower size data type which can result in loss of information.

Character functions

isLetter() Determines whether the specified char value is a letter.

isDigit() Determines whether the specified char value is a digit.

isWhitespace() Determines whether the specified char value is white space.

isUpperCase() Determines whether the specified char value is uppercase.

isLowerCase() Determines whether the specified char value is lowercase.

toUpperCase() Returns the uppercase form of the specified char value.

toLowerCase() Returns the lowercase form of the specified char value.

toString() Returns a String object representing the specified character value that is, a one character string.

Operators

- Assignment Operators
- Arithmetic Operators
- Relational Operators
- Bitwise Operators
- Logical Operators

Control Structures are just a way to specify flow of control in programs. There are three basic types of logic, or flow of control, known as:

- Sequence logic, or sequential flow
- Selection logic, or conditional flow
- Iteration logic, or repetitive flow

An **array** is a container object that holds a fixed number of sequential collection of elements of a same type.

2D array Represented in the form of rows and columns, also known as matrix. It is also known as array of arrays or list of arrays.

A method is a block of code or collection of statements or a set of code grouped together to perform a certain task or operation.

Why use methods?

- To reuse code: define the code once, and use it many times.
- Provides easy modification of the code, just by adding or removing a chunk of code.

The **method declaration** provides information about method attributes, such as visibility, return-type, name, and arguments.

Types of Methods

1. User defined Methods

- Written by the programmer according to the requirements.

2. Predefined Method

- Method that is already defined in the Java class libraries.
- Also known as the standard library method or built-in method.
- Eg: length(), equals(), compareTo(), sqrt(), etc.

Method Declaration

• **Return Type**: Return type is a data type that the method returns. It may have a primitive data type, object, collection, void, etc. If the method does not return anything, we use void keyword.

• **Method Name**: It is a unique name that is used to define the name of a method. It must be corresponding to the functionality of the method. Suppose, if we are creating a method for subtraction of two numbers, the method name must be subtraction(). A method is invoked by its name.

• **Parameter List**: It is the list of parameters separated by a comma and enclosed in the pair of parentheses. It contains the data type and variable name. If the method has no parameter, left the parentheses blank.

• **Method Body**: It is a part of the method declaration. It contains all the actions to be performed. It is enclosed within the pair of curly braces.

Access specifier or modifier is the access type of the method. It specifies the visibility of the method. Java provides four types of access specifier:

- **Public**: The method is accessible by all classes when we use public specifier in our application.
- **Private**: When we use a private access specifier, the method is accessible only in the classes in which it is defined.
- **Protected**: When we use protected access specifier, the method is accessible within the same package or subclasses in a different package.
- **Default**: When we do not use any access specifier in the method declaration, Java uses default access specifier by default. It is visible only from the same package only.

Static Method vs Instance Method

Static Method

- Method that belongs to a class rather than an instance of a class.
- Can call it without creating an object.
- It is invoked by using the class name.

Instance Method

- Method belongs to an instance of a class.
- Must create an object of the class to invoke it.
- It is invoked by using the object name.

Program Errors

• **Syntax errors (compile time)**

– Caused by an incorrectly written code such as misspelled keyword or variable, and incomplete code.

• **Runtime errors**

– Occurs when a statement attempts an operation that is impossible to carry out.

– Ex: division by zero.

• **Logical errors**

– Occurs when code compiles and runs without generating an error, but the result produced is incorrect.

An **exception** is an event, which occurs during the execution of a program, that disrupts the normal flow of the program's instructions.

Types of Java Exceptions

Checked exceptions - ArithmeticException, ClassCastException, NullPointerException.

Unchecked – Runtime exceptions, Errors.

Checked exceptions – Occurs at the compile time, Also called as compile time exceptions.

Exception handling in java is one of the powerful mechanisms to handle the runtime errors.

A **finally block** of code always executes, whether or not an exception has occurred.

Throwable

- Superclass of all errors and exceptions
- All exceptions and errors extend from a common java.lang.Throwable parent class.

Throws Keyword in the signature of a method to indicate that the method might throw one of the listed type exceptions.

This **lesson** provides the knowledge about Access Modifiers that specify the accessibility or scope of a field, method, constructor, or class.

Modifiers are Language Keywords that modify the definition of a Class, Method or a Variable.

Access modifiers are declared immediately before the type of a member variable or the return type of a method.

- There are four access levels :
 - Default
 - Public
 - Protected
 - Private

The **Default access modifier** is applied by default in the absence of an access modifier.

An **object** is a self-contained element of a computer program that represents a related group of features and is designed to accomplish specific tasks. EX- Employee-name, age, salary

A **Class** is a template used to create multiple objects with similar features.

Attributes Individual things that differentiate one class of objects from another and determine the appearance, state and other qualities of that class. E.g: Color orange, raw umber, lemon yellow, maize.

Instance variables and Class variables

• **Instance variables**

- An instance variable is an item of information that defines an attribute of one particular object.
- The object's class defines what kind of attribute it is, and each instance object stores its own value for that attribute.
- Instance variables also known as object variables.

• **Class variables**

- A Class variables is an item of information that defines an attribute of an entire class.
- The variable applies to the class itself and to all of its instances.

Behavior is the way that a class of objects can do to change their attributes, and also what they do when ask them to do something.

- Get angry – Calm down – Eat a peasant.

Structured

- Is the traditional approach of software development based upon the waterfall model that focus is only on process and procedures.

Object Oriented

- Whereas in Object Oriented, the focus is more on capturing the real world objects in the current scenario that are of importance to the system.

object-orientation, your program will be split into several small, manageable, reusable programs.

Attributes It's the characteristics or the properties of the Object

Methods Methods are the things that the Object can do

Encapsulation referrers to combining data and associated functions as a single unit.

Abstraction

- Hiding unnecessary details from the ordinary user.
- It handle complexity .

Inheritance

- Forming a new class from an existing class.
- Existing class is base class/parent class/super class and new class is derived class/child class/sub class.

Polymorphism

- Polymorphism is the state where an object can take the shape of many different forms, and lets us do the right thing at the right time.
- It describes the concept that objects of different types can be accessed through the same interface

Overriding When the method signature (name and parameters) are the same in the superclass and the child class, it's called overriding.

Overloading When two or more methods in the same class have the same name but different parameters, it's called overloading.

Components of a Language?

- Semantics - Meaning
- Syntax - How the meaning is presented

Java Virtual Machine?

Collection of software components that enables a computer to run a Java program.

Features of Java Language

- Case sensitive • Secure. • Robust
- Portable • Multithreaded

Integrated Development Environment

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Software versioning

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Array Length

<array_reference> .length will provide the length of the array object.

exception object

- When an error occurs within a method, the method creates an object and hands it off to the runtime system.
- The object, called an exception object.

Integrated Development Environment features

Code editor , debugging tool ,syntax highlight

Infinite loop

A loop keeps running forever without reaching a condition that would cause it to exit.

do while

The do-while loop first executes the code block and then evaluates the condition. If the condition is true, the loop continues; otherwise, it exits.

for

Provides a compact syntax to specify the initialization, condition, and increment/decrement in a single line by specifying the exact number of iterations in the loop declaration.

while

Loop evaluates the condition before each iteration. If the condition is true, the loop continues; otherwise, it exits.

Purpose of word

break

Used to terminate the innermost loop or switch statement.

continue

Used to skip the remaining code within the loop for the current iteration and move to the next iteration.

Java swing framework

Power full set of gui components and tools that are used for create desktop application in java.

Event handling and step

Process of capturing and responding to various event triggers by user interactions or system actions .

Step 1

Implement the listener interface

Step 2

Register an event listener for the component

Step 3

implement the event handling code in side the listener method.